

2nd CIRP CSI - 29/05/2014

INFLUENCE OF CUTTING PROCESS MECHANICS ON SURFACE INTEGRITY AND ELECTROCHEMICAL BEHAVIOR OF OFHC COPPER

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Outline



Introduction



Objectives



Experimental procedure and parameters

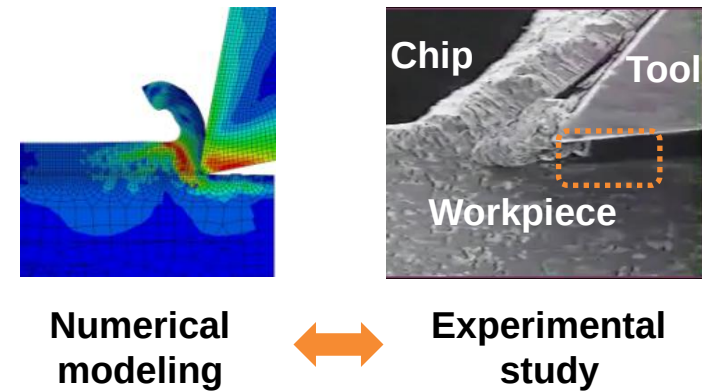
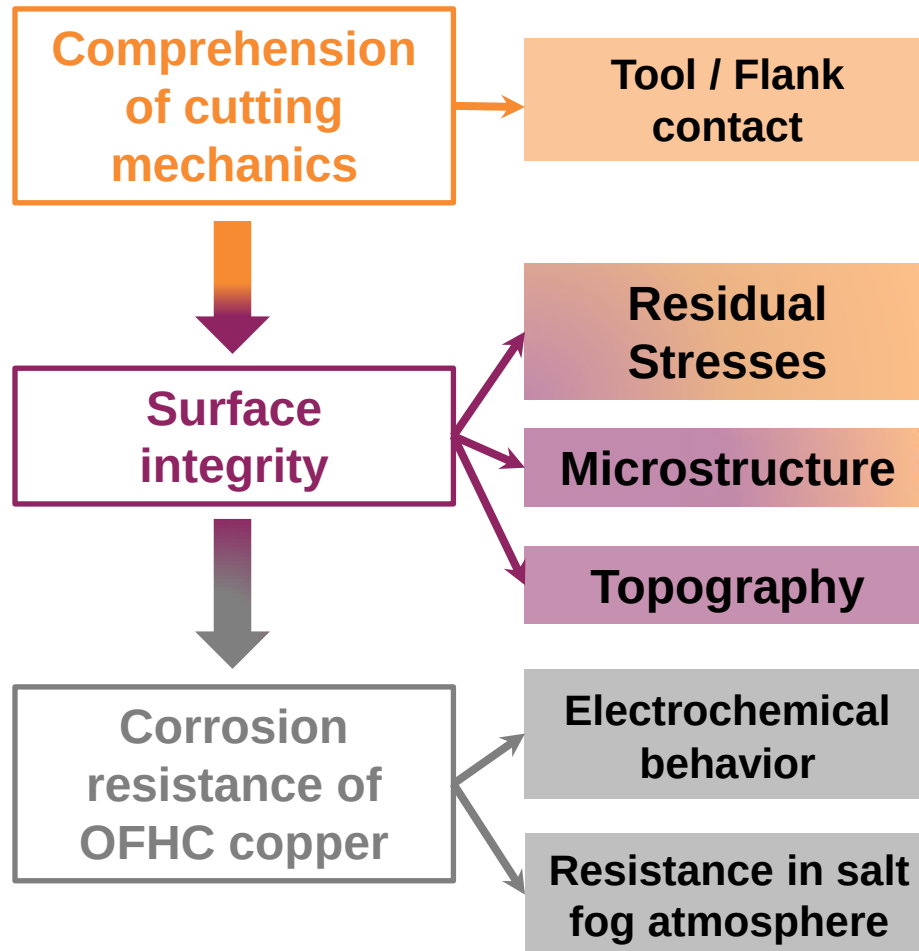


Results and discussion

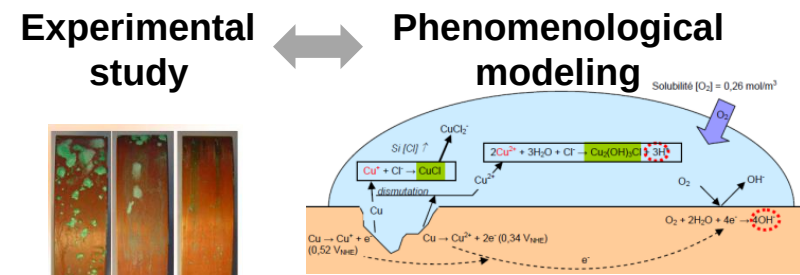
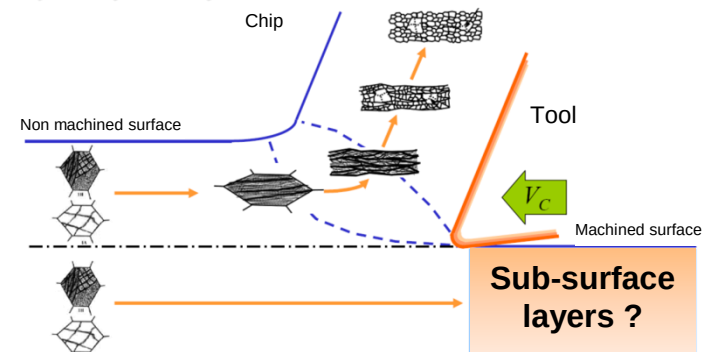


Conclusion and outlook

Introduction



F, T°, RS, DRX

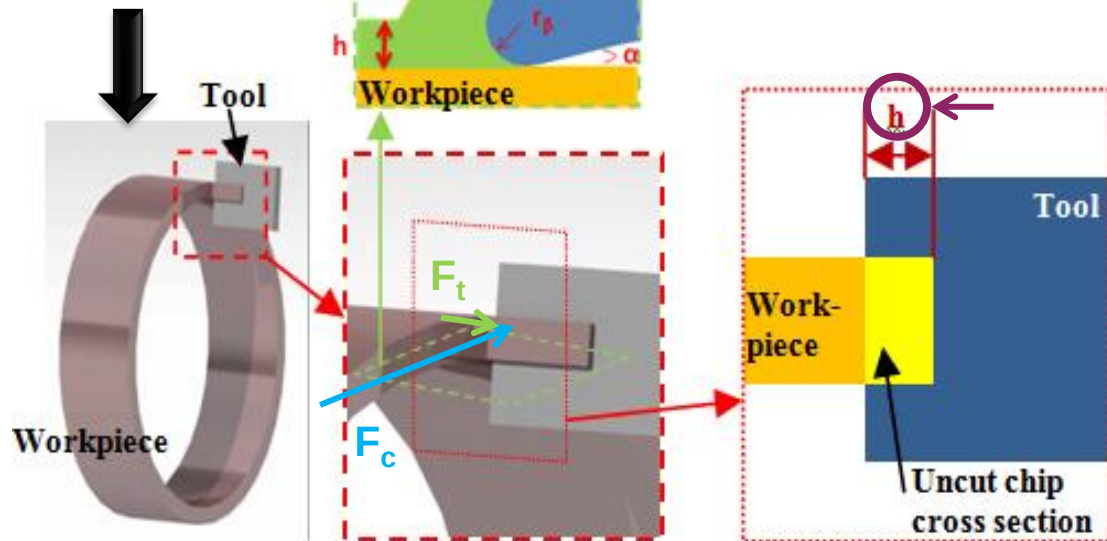


Objectives

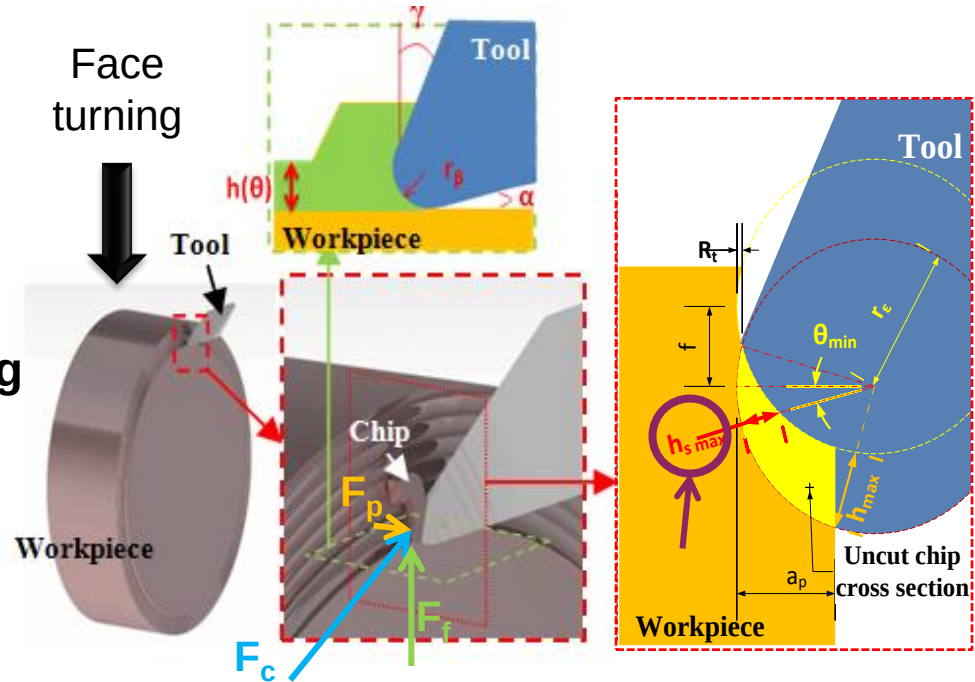
Compare a typical 3D turning with orthogonal cutting

- Influence of superfinishing machining conditions on surface integrity and corrosion resistance of OFHC copper

Orthogonal cutting



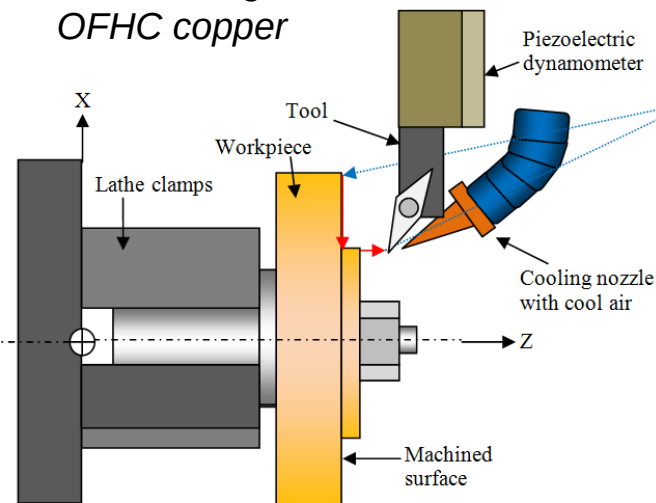
Face turning



Is orthogonal cutting process able to provide a surface integrity similar to that one generated by 3D cutting?

Experimental procedure and parameters

Face turning of
OFHC copper



$f = 0.1 ; 0.15 ; 0.2 \text{ mm/rev}$

$a_p = 0.15 ; 0.30 ; 0.50 \text{ mm}$

$V_c = 120 \text{ m/min}$ Air cooling

$-5 \pm 1^\circ\text{C}$

$\alpha = 7^\circ$

$\gamma = 20^\circ$

$r_\beta = 9 \mu\text{m}$

1M NaClO₄ 1 micro-capillary

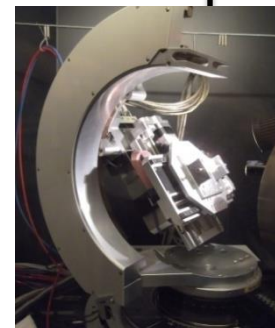
3D Face turning

Forces

Topography

RS

Local
electro-
chemical
tests



Polarization

E_{piq} I_{pass} E_{cor}

Electrochemical
behavior

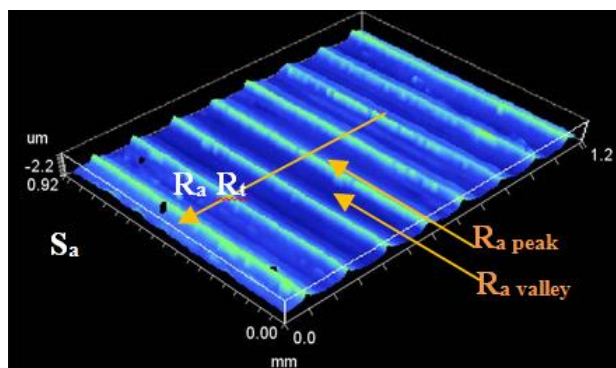
F_c, F_p, F_f, K_c

Cutting
mechanics

R_t, R_a

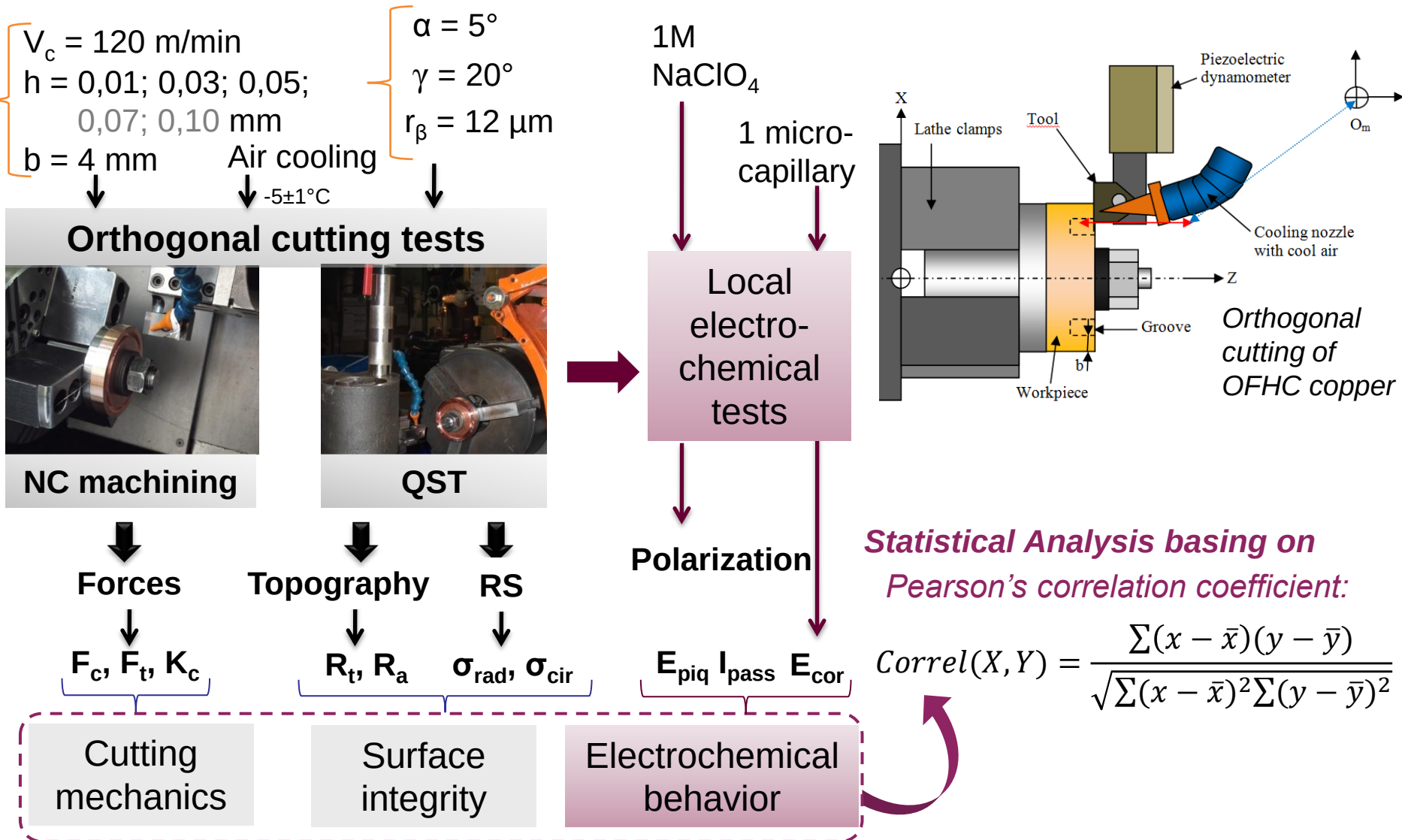
Surface
integrity

$\sigma_{rad}, \sigma_{cir}$

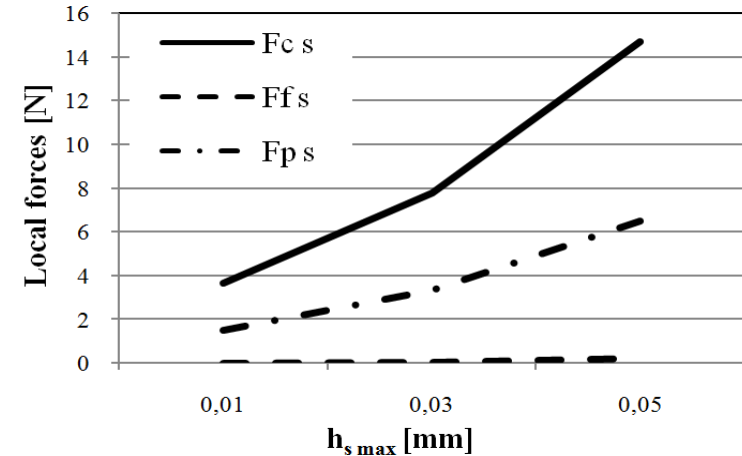
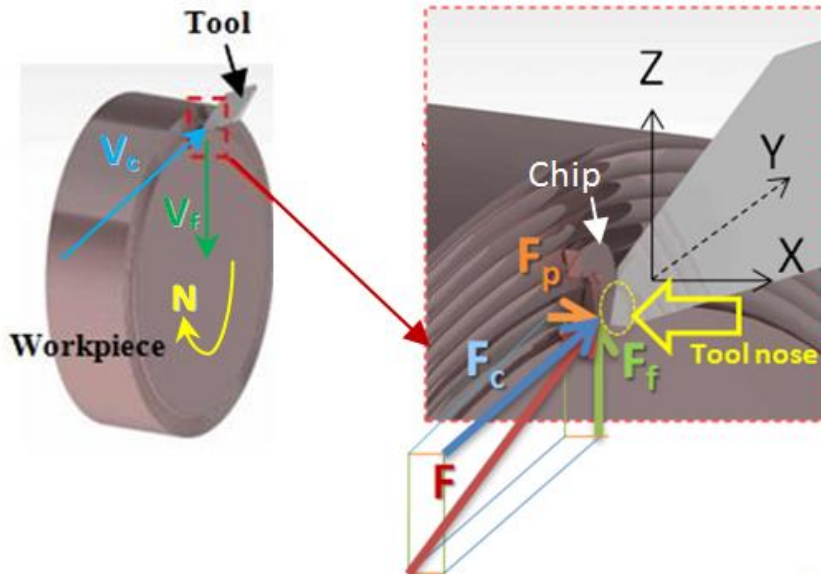


Roughness measurement
directions

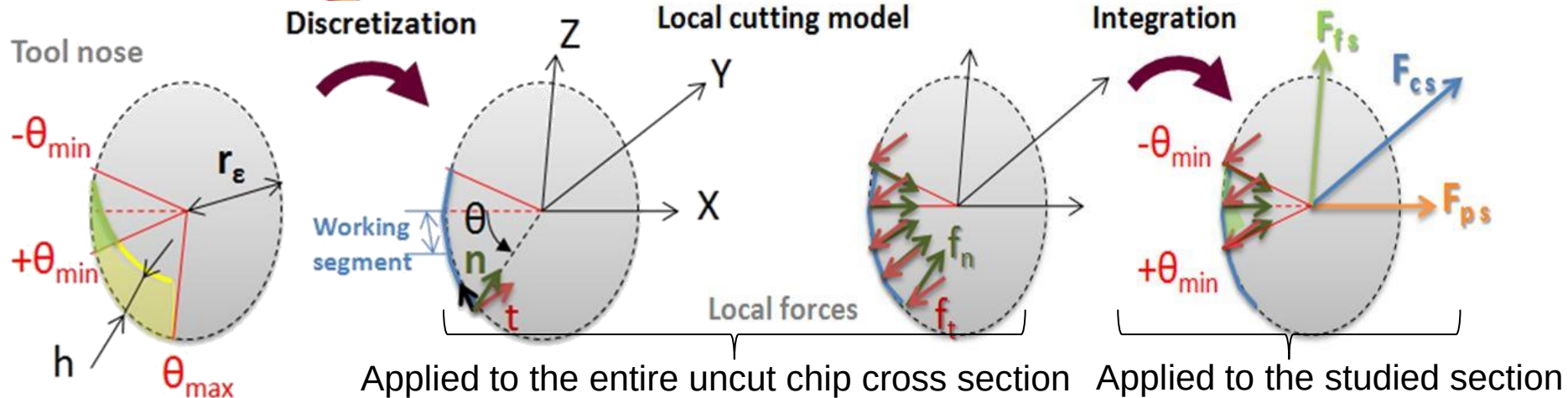
Experimental procedure and parameters



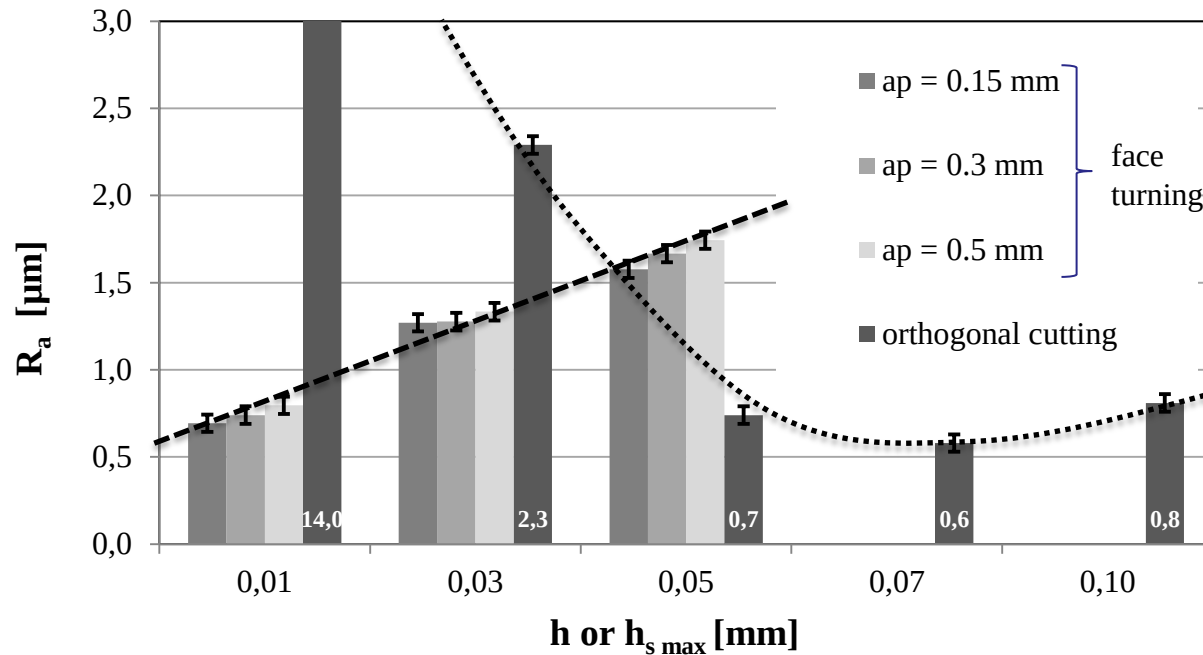
Results and discussion: Cutting mechanics



Local forces in face turning [$f(h_{s \max})$]



Results and discussion: Surface integrity



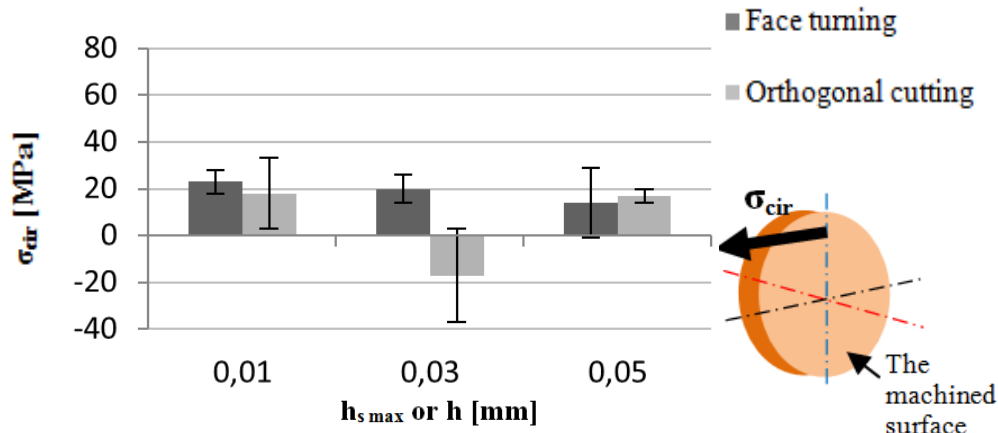
R_a in face turning [$f(h_{s \max}, a_p)$] and in orthogonal cutting [$f(h)$]

- $h_{s \max}$ is correlated to R_a , R_t and S_a by **> 97%**.
- The influence of h on the surface roughness is **opposite** to the **orthogonal cutting**.

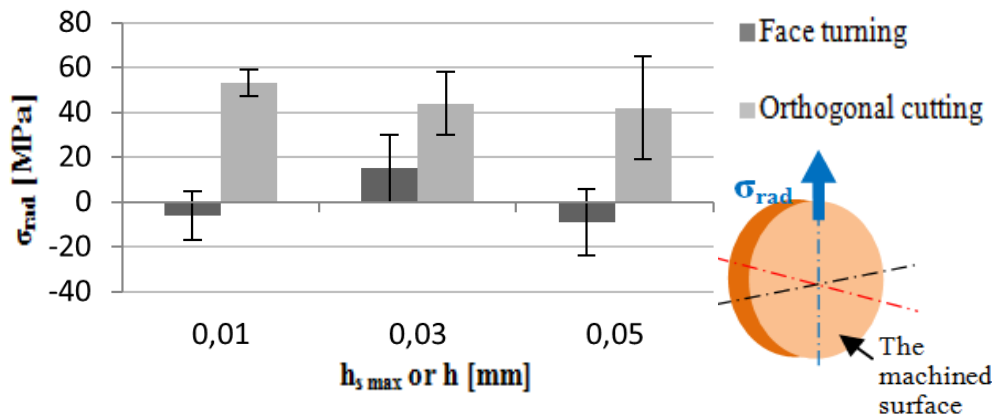
➡ Cutting instability for very low h/r_β ratios and very large b .

- $h > 0.05$ mm : R_a tends to a **steady state** and depends only of the tool wear.

Results and discussion: Surface integrity



σ_{cir} at samples' surfaces



σ_{rad} at samples' surfaces

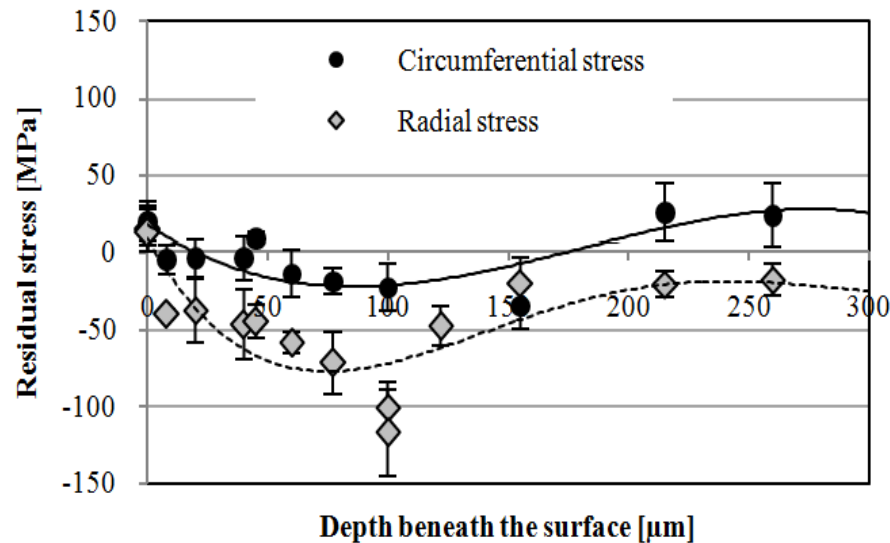
- No significant influence of $h_{s max}$ or h on RS

- Surface RS are **tensile** for face turning and **orthogonal cutting**.

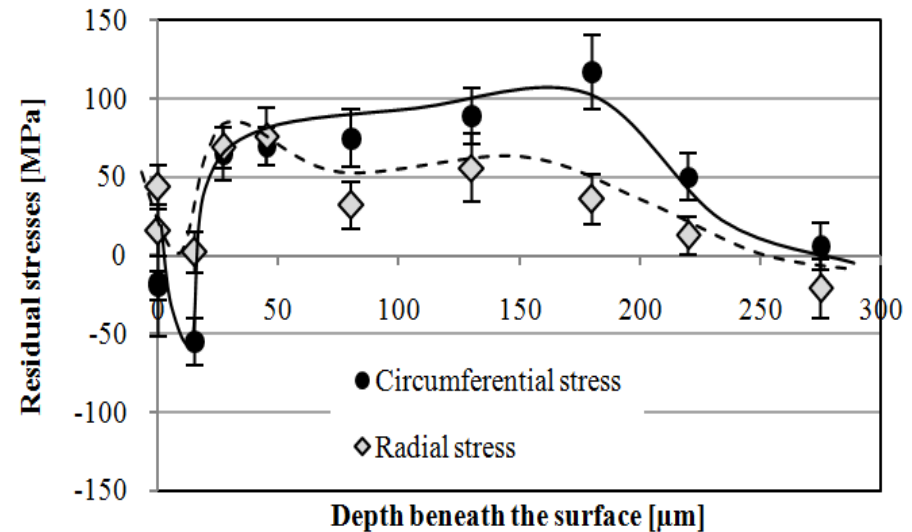
- Surface σ_{rad} in **orthogonal cutting** is **higher** than that in **face turning** (~ 0)

- Local forces are **inversely correlated** with surface σ_{cir} [face turning : **>99%**], as well as with the surface σ_{rad} in **orthogonal cutting** (**>74%**).

Results and discussion: Surface integrity



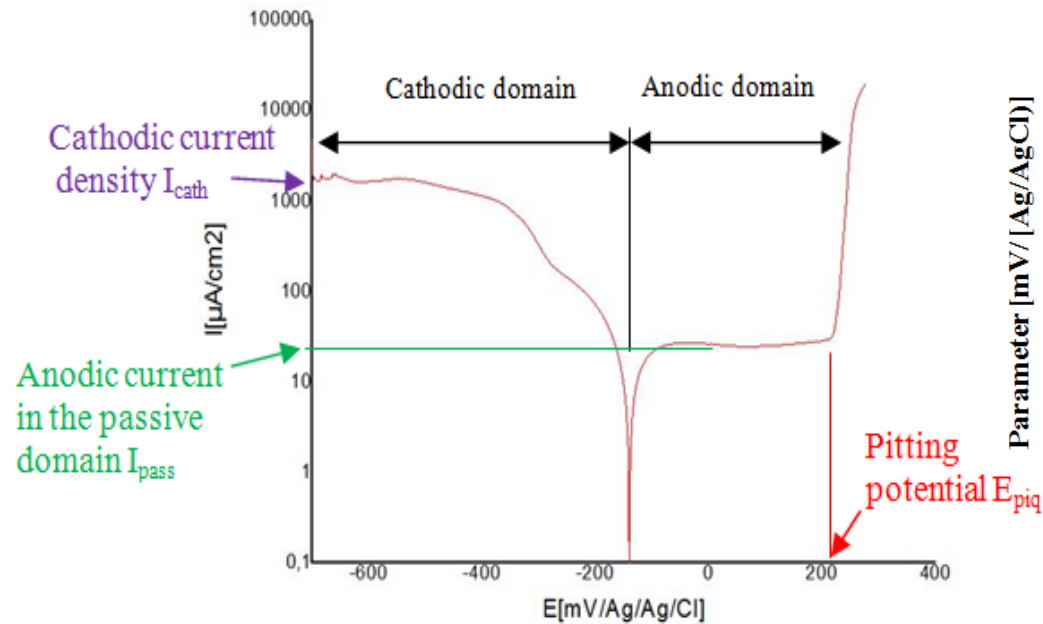
In depth RS profiles
[face turning; $h_{s \max} = 0.03 \text{ mm}$]



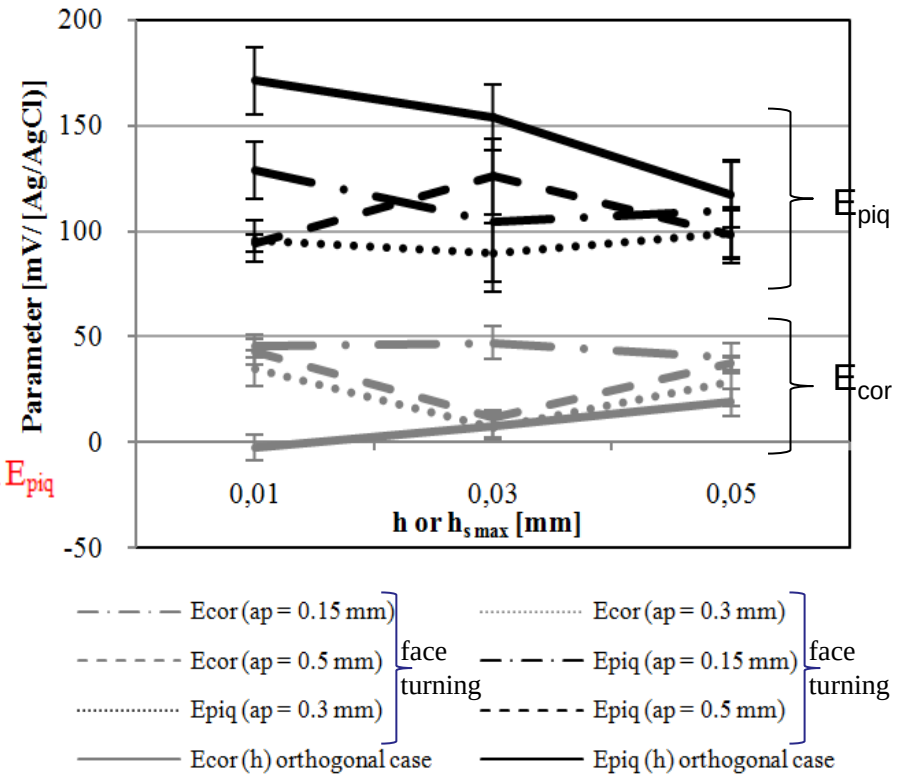
In depth RS profiles
[orthogonal cutting; $h = 0.03 \text{ mm}$]

- Below the surface, **compressive** stresses are generated by **face turning**, while **orthogonal cutting** generates **tensile** stresses for depth greater than 20 μm .

Results and discussion: Local electrochemical behavior

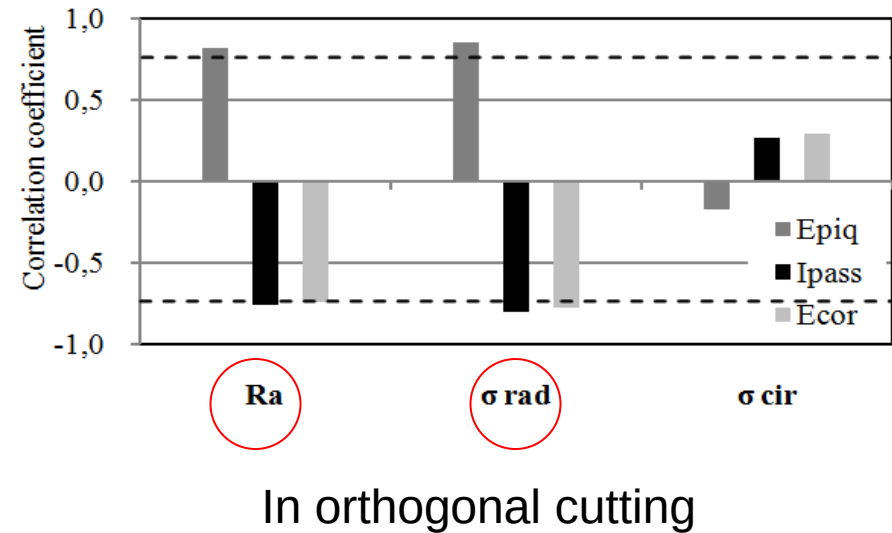
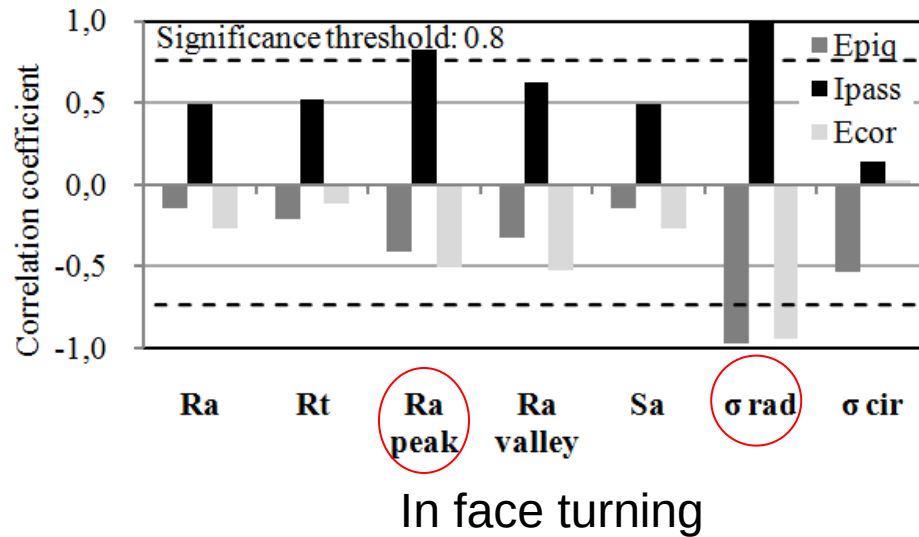


Polarization curve composition



E_{cor} and E_{piq} evolution [$f(h \text{ or } h_{\text{s max}})$]

Results and discussion: Correlations between SI parameters and electrochemical reactivity parameters



• σ_{rad} , R_{a_peak} (face turning) and R_a (orthogonal cutting) are the parameters influencing significantly the local electrochemical reactivity.

Conclusion and Outlook

- $h_{s\ max}$ (face turning) and h (orthogonal cutting) are strongly correlated to the local forces and surface roughness, but not to the surface residual stresses.
- Concerning to the in-depth residual stress profiles, face turning generates a thicker layer having compressive residual stresses, while orthogonal cutting generates a thicker layer having tensile residual stresses.
- Correlation analysis has proven that R_{a_peak} (face turning) and R_a (orthogonal cutting) are the most influencing parameters on the local electrochemical reactivity.

Conclusion and Outlook

The present results are not enough to confirm the hypothesis that identical deformation process is applied to generate the machined surface in both superfinishing turning and orthogonal cutting.



Further experiments are required with closer analysis to the **thermal** and **mechanical** phenomena developed at the **tool flank contact**.

Thanks for listening

